MARCH 23, 2026

Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect with 20 years of experience designing large-scale distributed platforms and cross-platform data infrastructure. Expert at translating evolving standards into production systems adopted across hundreds of enterprise environments. Known for unifying complex multi-system integrations into scalable architectures spanning APIs, cloud services, and AI-enabled workflows.

Ask Me || DM me on LinkedIn

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Madison, WI

linkedin.com/in/matt-varghese

github.com/mattvarghese

WORK EXPERIENCE

Software Systems Architect, Health Plans

Epic - Verona, WI — *MARCH 2021 - PRESENT*

Architect for a large-scale, standards-based integration platform, defining automated real-time decision workflows for enterprise-scale organizations. Engineered systems adopted by hundreds of major enterprise clients, reducing high-latency business process turnarounds from several days to near real-time.

- **Led architecture and delivery of an enterprise interoperability platform** for real-time data exchange, collaborating with standards bodies and external vendors to transform emerging specifications into scalable, production-ready infrastructure.
- **Identified gaps and established a unified "source of truth"** replacing fragmented logic across 12+ product modules, reducing lead time for new feature development by ~80%, and improving long-term maintainability.
- **Recognized the need and architected standardized** data models and execution frameworks replacing inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized third-party decision-engine integrations** using OAuth2 SSO patterns, enabling real-time workflow execution and reducing integration defects by ~60%.
- **Mitigated tooling deficiencies by building simulation and testing frameworks** to emulate third-party systems, enabling end-to-end workflow validation, proactively eliminating defects, and accelerating partner onboarding.

Senior Software Systems Developer, Clinical Informatics

Epic - Verona, WI — *AUGUST 2010 - MARCH 2021*

Developed clinical decision support (CDS) integrations, web app integrations, and mobile push notifications. Directed adoption of modern healthcare interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Created Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

SKILLS

Architecture & Infrastructure

- SDLC
- Distributed Systems
- Microservices
- Azure
- CI/CD
- Docker
- Virtualization
- Hadoop
- Maintainable Architecture

Engineering Stack

- TypeScript
- React
- Web API
- Node
- .NET/C#
- C/C++
- Python
- Git & SVN
- ANSI-M
- SQL & NoSQL
- Linux & Windows

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Network Architecture & Routing

COLLABORATION

Standards-body Hackathons

Developed prototypes of industry specifications in collaboration with participants from other vendors to improve interoperability.

Prior Auth Burden Reduction

Represented Epic on public calls steering development of standards to reduce admin burden in Prior Authorization.

- **Identified fragmentation and led enterprise-wide centralization** of 3rd-party web application integrations to establish standards for security and maintainability.
- **Designed & Developed Epic's decision support web services framework,** enabling standards-based clinical decision support across multiple clinical applications.
- **Identified usability gaps and redesigned** decision support configuration to align with mental models, improving usability and enabling efficacy feedback loops for doctors.
- **Recognized subject matter expert in Web App Security and RESTful integrations,** providing architectural guidance to multiple teams and onboarding external vendors.

KEY PROJECTS

AI Assistant for patients and doctors
2025 Q2

Integrates LLM APIs with electronic patient health record for agentic use-cases. LLM chat web app answers doctor and patient questions using real health data. Enables AI driven forms, AI decision support, AI care delivery, etc.

JSON Serializer & Deserializer for ANSI-M
2020

Spotlights long-term vision in creating powerful generic reusable tooling. Created RFC 8259 compliant JSON Serializer and Deserializer for ANSI-M. Serves as a foundation for 30+ database AI integrations across Epic.

MIPS Pipeline Simulator
2005, 2025

github.com/mattvarghese/coconut

Demonstrates deep expertise in low-level systems, architecture, parallelism, and performance. 5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005. In 2025, developed a Small-C compiler above the assembler with AI assistance.

PRIOR EXPERIENCE

Research Assistant, Condor Distributed Computing
University of Wisconsin - Madison, WI — *SEPT 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing
Cisco Systems - Bangalore, India — *JUNE 2006 - JULY 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

EDUCATION

Master of Science, Computer Science
University of Wisconsin - Madison — *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering
National Institute of Technology - Calicut, India — *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

Authoring Industry Standards
Contributed to HL7 CDS Hooks and HL7 SMART Web Messaging specifications to enhance workflow integration and security.

Epic Vendor Conferences
Presented for several years on Epic's web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring
Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment
Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

EXPLORATORY

Web Based SSH Client
2025 Q3

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell access to servers.

Demonstrated proof of concept fully browser-based SSH client for unix servers.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder
2024 Q4

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling